

HARNESSING WIND POWER

WIND TURBINES

Wind power, along with other sources such as solar power and hydropower (power from water), is a form of renewable energy—an energy source that never runs out. The Wind Tree featured here operates on a small, local level but harnesses the wind in a similar way to the giant turbines found on wind farms (see pages 46–47). Wind turbines turn the power of the wind into electrical power. The blades of a turbine rotate when the wind blows, turning a shaft and driving a machine called a generator to convert the mechanical energy of this motion into electrical energy.

MICROGENERATION

Wind turbines come in a range of sizes and designs. Mini turbines can be mounted on rooftops to generate small amounts of electricity either to meet the energy needs of a household or business or to add to the power grid.



GENERATORS

When the wind pushes on the turbines, its energy is turned into mechanical energy. This then passes to a generator, which is a machine that converts mechanical energy into electrical energy. It works using magnets. The energy from the blades spins a coil of wire around inside the magnet. This generates a flow of electrons—an electric current (see page 34). This current passes through cables to the power grid.

